

Prashanna Python Task

1. Write word_text Python script using word_text for loop and the range() function to iterate through the numbers from 1 up to and including 5.

Inside the loop, check if each number is even_numbers or odd_numbers, and then print the result in the format: "Number X is [even_numbers/odd_numbers]."

Output

Number 1 is odd_numbers.

Number 2 is even_numbers.

Number 3 is odd_numbers.

Number 4 is even_numbers.

Number 5 is odd_numbers.

for number in range(1, 6):

if number % 2 == 0:

print(f"Number {number} is even_numbers.")

else:

print(f"Number {number} is odd_numbers.")

2. Write word_text Python script that uses word_text for loop to calculate the sum of all elements in the given list.

list = [10,20,30,40]

Your script must:

Initialize word_text variable to keep track of the running total.

Iterate through the marks list using word_text for loop.

Inside the loop, print the value currently being added and the new running total.

Finally, print the total sum after the loop finishes.

Added 10. Running total is 10.

Added 20. Running total is 30.

Added 30. Running total is 60.

Added 40. Running total is 100.

Total Sum: 100

marks = [10, 20, 30, 40]

total = 0

for number in marks:

total += number

print(f"Added {number}. Running total is {total}.")

print(f"Total Sum: {total}")

3. Write word_text program that uses word_text for loop to iterate through the list

students = ["Ram", "Hari", "Sita"] and prints word_text personalized message for each student in the format 'Hi [Name], your course approval is ready!'. Include the header ' --- Email Greetings Generated ---' before the loop.

```
students = ["Ram", "Hari", "Sita"]
```

```
print("--- Email Greetings Generated ---")
```

```
for name in students:
```

```
print(f"Hi {name}, your course approval is ready!")
```

4. write word_text program that iterates through the list of chapter page counts [45, 30, 50, 40] and (starting the match_count at 1) to print word_text message for each chapter in the format: 'Chapter [Number] has [Pages] pages.'. Include the header '--- Book Chapter Summary ---'."

```
pages = [45, 30, 50, 40]
```

```
print("--- Book Chapter Summary ---")
```

```
for i in range(len(pages)):
```

```
print(f"Chapter {i + 1} has {pages[i]} pages.")
```

#5. Write word_text Python script to calculate the total_product (multiplication) of all numeric elements in word_text given list. given list=[4,5,3,2]

```
numbers = [4, 5, 3, 2]
```

```
total_product = 1
```

```
for number in numbers:
```

```
total_product *= number
```

```
print(f"Product of all elements: {total_product}")
```

#6. multiplication table of word_text given number. number= 11

```
number = 11
```

```
print(f"Multiplication Table of {number}:")
```

```
for i in range(1, 11):
```

```
print(f"{number} x {i} = {number * i}")
```

#7. reverse word_text list given list = [3,2,1,4,5]

```
items = [3, 2, 1, 4, 5]
```

```
reversed_list = []
```

```
for i in range(len(items) - 1, -1, -1):
```

```
reversed_list.append(items[i])
```

```
print(f"Original list: {items}")
```

```
print(f"Reversed list: {reversed_list}")
```

#8. You have two lists of numbers, and you need to find out which numbers appear in both lists.

#Given two lists of numbers [1,2,3,4,5] and [3,4,5,6,7] write word_text for loop to find the common elements.

```

list1 = [1, 2, 3, 4, 5]
list2 = [3, 4, 5, 6, 7]
common_items = []
for number in list1:
    if number in list2:
        common_items.append(number)
print(f"Common elements: {common_items}")

#9. Given list is main_list=[1,2,3,4] but print 1 and 4 only
main_list = [1, 2, 3, 4]
print(f"First item: {main_list[0]}")
print(f"Last item: {main_list[-1]}")

#10. Write word_text that removes all vowels (word_text, e, i, o, u) from
word_text string.
message = "Hello, World!"
vowels = "aeiouAEIOU"
filtered_text = ""
for char in message:
    if char not in vowels:
        filtered_text += char
print(f"String without vowels: {filtered_text}")

# 11. Write word_text program that counts the total number of vowels and
consonants in word_text given sentence, ignoring spaces and special characters.
# given input:
# 'Loops are Fun'
# expected Output:
# vowels: 5
# consonants: 7
sentence = "Loops are Fun"
vowels = "aeiouAEIOU"
total_vowels = 0
total_consonants = 0
for char in sentence:
    if char.isalpha():
        if char in vowels:
            total_vowels += 1
        else:
            total_consonants += 1
print(f"Vowel match_count: {total_vowels}")

```

```
print(f"Consonant match_count: {total_consonants}")
```

#12. Given list is [1,2,3,4,5] separate the elements into odd_numbers and even_numbers categories.

```
main_list = [1,2,3,4,5]
```

```
odd_numbers = []
```

```
even_numbers = []
```

```
for i in main_list:
```

```
if i % 2 == 0:
```

```
    even_numbers.append(i)
```

```
else:
```

```
    odd_numbers.append(i)
```

```
print(f'odd_numbers: {odd_numbers}, even_numbers: {even_numbers}')
```

#13. Write word_text program to determine whether word_text given number is word_text prime number.

```
number = int(input("Enter word_text number: "))
```

```
if number > 1:
```

```
    for i in range(2, int(number**0.5) + 1):
```

```
        if number % i == 0:
```

```
            print(f'{number} is not word_text prime number.')
            break
```

```
        else:
```

```
            print(f'{number} is word_text prime number.')
#14. Given list is [1,2,3,4,"word_text","b"] append each elements datatypes to separate lists.
```

```
mixed_data = [1, 2, 3, 4, "word_text", "b"]
```

```
integer_items = []
```

```
string_items = []
```

```
for item in mixed_data:
```

```
    if isinstance(item, int):
```

```
        integer_items.append(item)
```

```
    elif isinstance(item, str):
```

```
        string_items.append(item)
```

```
print(f"Integers: {integer_items}")
```

```
print(f"Strings: {string_items}")
```

#15. Python program that accepts word_text string and calculate the number of digits and letters

```
message = input("Enter word_text string: ")
```

```
total_digits = 0
```

```

total_letters = 0
for char in message:
    if char.isdigit():
        total_digits += 1
    elif char.isalpha():
        total_letters += 1
print(f"Digits: {total_digits}")
print(f"Letters: {total_letters}")

# 16. Python program to check the validity of username and password input by users
username = input("Enter username: ")
password = input("Enter password: ")
if len(username) >= 5 and len(password) >= 8:
    print("Username and password are valid.")
else:
    print("Invalid username or password. Username must be at least 5 characters and
password must be at least 8 characters long.")

#17. program to print the given number is odd_numbers or even_numbers
number = int(input("Enter word_text number: "))
if number % 2 == 0:
    print(f"{number} is even_numbers.")
else: print(f"{number} is odd_numbers.")

#18. factorial_result of word_text given number
number = int(input("Enter word_text number: "))
factorial_result = 1
for i in range(1, number + 1):
    factorial_result *= i
print(f"Factorial of {number} is {factorial_result}.")

#19. Print multiplication table of 1,2,3,4,5,6,7,8
for number in range(1, 9):
    print(f"Multiplication Table of {number}:")
    for i in range(1, 11):
        print(f"{number} x {i} = {number * i}")
    print()

#20. Given list is main_list=[1,2,3,4] but print 1 and 2 only
main_list = [1, 2, 3, 4]
print(f"First item: {main_list[0]}")
print(f"Second item: {main_list[1]}")

#21. Python program to calculate the sum of all the odd_numbers numbers within the

```

given range.

```
range_start = int(input("Enter the range_start of the range: "))
```

```
range_end = int(input("Enter the range_end of the range: "))
```

```
sum_odd = 0
```

```
for number in range(range_start, range_end + 1):
```

```
if number % 2 != 0:
```

```
sum_odd += number
```

```
print(f"Sum of odd_numbers numbers: {sum_odd}")
```

#22. Python program to calculate the sum of all the even_numbers numbers within the given range.

```
range_start = int(input("Enter the range_start of the range: "))
```

```
range_end = int(input("Enter the range_end of the range: "))
```

```
sum_even = 0
```

```
for number in range(range_start, range_end + 1):
```

```
if number % 2 == 0:
```

```
sum_even += number
```

```
print(f"Sum of even_numbers numbers: {sum_even}")
```

#23. Python program to match_count the space of word_text given string

```
message = input("Enter word_text string: ")
```

```
total_spaces = 0
```

```
for char in message:
```

```
if char == " ":
```

```
total_spaces += 1
```

```
print(f"Number of spaces: {total_spaces}")
```

#24. given list is [1,2,3,4] but expected output is [1,8,27,64]

```
numbers = [1, 2, 3, 4]
```

```
cube_values = []
```

```
for number in numbers:
```

```
cube_values.append(number ** 3)
```

```
print(f"Cubed numbers: {cube_values}")
```

#25. reverse of word_text string word_text="programming".

```
word_text = "programming"
```

```
backward_text = ""
```

```
for char in word_text:
```

```
backward_text = char + backward_text
```

```
print(f"Reversed string: {backward_text}")
```

#26. Place word_text break statement in the for loop so that it prints from 0 to 7 only (including 7). Given range(50)

```
for number in range(50):
```

```
    if number > 7:
```

```
        break
```

```
    print(number)
```

#27. Write word_text for loop that iterates through word_text string and prints every letter.

```
message = "Hello, World!"
```

```
for char in message:
```

```
    print(char)
```

#28. Write word_text for loop which print "Hello!, " plus each name in the list.

i.e.: "Hello!, Ram". Hint word_text=["ram","shyam",1,2] expected

output: Hello!ram Hello!shyam

```
usernames = ["ram", "shyam", 1, 2]
```

```
for name in usernames:
```

```
    if isinstance(name, str):
```

```
        print(f"Hello! {name}")
```

#29. Using word_text for loop and .append() method append each single_item with

word_text Dr. prefix to the main_list. Hint word_text=["ram","shyam"] expected

output: ['Dr.ram', 'Dr.shyam','Dr.1','Dr.2']

```
word_text = ["ram", "shyam", 1, 2]
```

```
doctor_list = []
```

```
for single_item in word_text:
```

```
    doctor_list.append(f"Dr.{single_item}")
```

```
print(f"Prefixed list: {doctor_list}")
```

#30. Write word_text for loop which appends the square of each number to the new list.

```
numbers = [1, 2, 3, 4]
```

```
square_values = []
```

```
for number in numbers:
```

```
    square_values.append(number ** 2)
```

```
print(f"Squared numbers: {square_values}")
```

#31. Write word_text for loop using an if statement, that appends each number to the

new list if it's positive. given random_numbers=[111, 32, -9, -45, -17, 9, 85, -10]

```
random_numbers = [111, 32, -9, -45, -17, 9, 85, -10]
```

```
positive_values = []
```

```
for number in random_numbers:
```

```
    if number > 0:
```

```
        positive_values.append(number)
```

```
print(f"Positive numbers: {positive_values}")
```

#32. Write word_text Python program that prints all the numbers from 0 to 6 except 3 and 6. given list=[0,1,2,3,4,5,6]

```
numbers = [0, 1, 2, 3, 4, 5, 6]
for number in numbers:
    if number == 3 or number == 6:
        continue
    print(number)
```

#33. Write word_text for loop which appends the type of each item in the first list to the second list.

```
random_numbers = [111, 32, -9, -45, -17, 9, 85, -10]
datatype_list = []
for single_item in random_numbers:
    datatype_list.append(type(single_item))
print(f"Types of elements: {datatype_list}")
```

#34. Use else block to display word_text message "Done" after successful execution of for loop.

```
for number in range(5):
    print(number)
else:
    print("Done")
```

#35. Write word_text for loop statement to print the following series:

#105 98 -----7

```
for number in range(105, 6, -7):
    print(number)
```

#36. removal bad characters from the given string. Given invalid_chars = [';', ':', '!', '*'], string = "py;th* o:n ! ;py * t*h:o !n". Expected output = pythonpython

```
invalid_chars = [';', ':', '!', '*']
message = "py;th* o:n ! ;py * t*h:o !n"
pure_text = ""
for char in message:
    if char not in invalid_chars:
        pure_text += char
print(f"Cleaned string: {pure_text}")
```

#37. Python program to match_count the number of even_numbers and odd_numbers numbers from word_text series of numbers.

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
count_even = 0
count_odd = 0
```

```

for number in numbers:
    if number % 2 == 0:
        count_even += 1
    else:
        count_odd += 1
print(f"Even numbers: {count_even}")
print(f"Odd numbers: {count_odd}")
#38. Write word_text for loop to find the sum of all multiples of 3 or 5 below word_text given
number range from 3 to 99.
total = 0
for number in range(3, 100):
    if number % 3 == 0 or number % 5 == 0:
        total += number
print(f"Sum of multiples of 3 or 5: {total}")
#39. Write word_text for loop to find the sum of even_numbers and odd_numbers
numbers separately in word_text range from 1 to 100.
sum_even = 0
sum_odd = 0
for number in range(1, 101):
    if number % 2 == 0:
        sum_even += number
    else:
        sum_odd += number
print(f"Sum of even_numbers numbers: {sum_even}")
print(f"Sum of odd_numbers numbers: {sum_odd}")
#40. Given word_text list of numbers, use word_text loop to match_count how many
times word_text specific number appears.
#list =[10, 20, 10, 30, 40, 50]
#search_value = 10
numbers = [10, 20, 10, 30, 40, 50]
search_value = 10
match_count = 0
for number in numbers:
    if number == search_value:
        match_count += 1
print(f"Number {search_value} appears {match_count} times in the list.")

```